Kinematics

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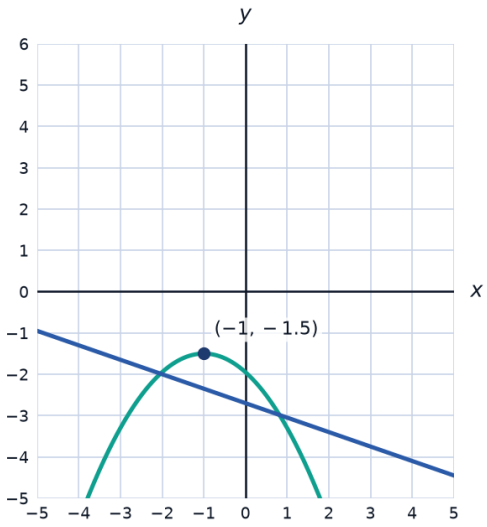
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Displacement, velocity, and acceleration**

A particle moves with displacement $s = t^3 - 6t^2 + 9t$ (in meters, t in seconds). Find expressions for its velocity v and acceleration a.

2 **[Foundational]** • **Target: Displacement, velocity, and acceleration**

A particle moves with displacement $s = 2t^3 - 9t^2 + 12t$ (in meters, t in seconds). Find its velocity at $t = 2$ seconds.



3 **[Foundational]** • **Target: The suvat equations for constant acceleration**

A car starts from rest and accelerates uniformly at 2.5 m/s^2 for 8 seconds. Find its final velocity and the distance traveled, using $v = u + at$ and $s = ut + \frac{1}{2}at^2$.

4 **[Foundational]** • **Target: The suvat equations for constant acceleration**

A ball is thrown upward with initial velocity $u = 15$ m/s. Using $g = 9.8$ m/s^2 (deceleration), find the maximum height reached, using $v^2 = u^2 + 2as$ with $v = 0$ at the top.

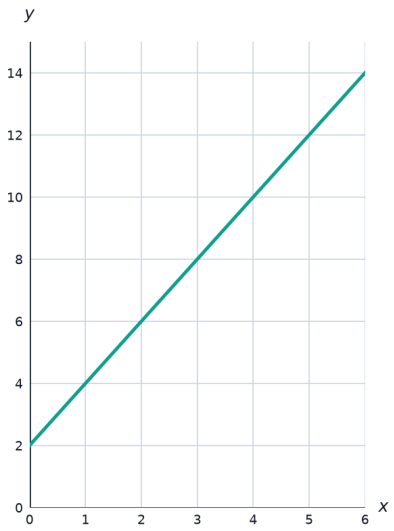
5 **[Foundational]** • **Target: The suvat equations for constant acceleration**

A cyclist decelerates uniformly from 12 m/s to rest over a distance of 36 meters. Find the deceleration, using $v^2 = u^2 + 2as$.

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6 [Foundational] • Target: Velocity-time graphs

A particle accelerates uniformly from $u = 2$ m/s to $v = 14$ m/s over 6 seconds, then travels at constant velocity for a further 4 seconds. The velocity-time graph is shown. Find the total distance traveled, using the area under the graph.



7 [Foundational] • Target: Velocity-time graphs

A particle decelerates uniformly from $v = 20$ m/s to rest over 5 seconds, giving a velocity-time graph that is a straight line from $(0, 20)$ to $(5, 0)$. Find the distance traveled, using the area under the graph.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Motion under gravity

A stone is dropped from a height of 45 meters. Using $g = 9.8$ m/s^2, find the time taken to reach the ground, and its speed on impact.

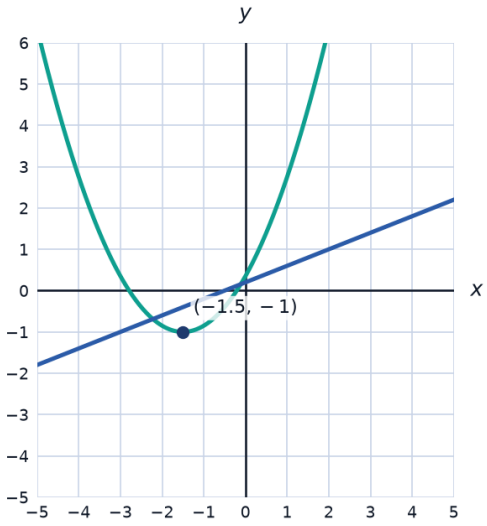
9 [Intermediate] • Target: Motion under gravity

A ball is thrown vertically upward with initial speed $u = 24.5$ m/s. Using $g = 9.8$ m/s^2, find the total time taken to return to its starting point.

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10 [Intermediate] • Target: Using calculus for variable acceleration

A particle has velocity $v = 3t^2 - 2t$ (m/s). Find its displacement from $t = 0$ to $t = 3$, and its acceleration at $t = 3$.



11 [Intermediate] • Target: Finding when a particle is at rest or changes direction

A particle has velocity $v = t^2 - 5t + 6$ (m/s). Find the times at which the particle is instantaneously at rest.

12 [Intermediate] • Target: Relative and combined motion

Two cars start 300 meters apart on a straight road and travel toward each other, one at 18 m/s and the other at 12 m/s, both at constant speed. Find the time until they meet.

13 [Intermediate] • Target: Relative and combined motion

Two cars start at the same point on a straight road, one traveling at 25 m/s. The second car starts 4 seconds later, traveling at 30 m/s. Find how long after the first car starts the second car catches up.

14 [Intermediate] • Target: A well-designed word problem: a train between two stations

This question has two parts. A train starts from rest at station A and accelerates uniformly at 0.5 m/s^2 until it reaches a speed of 20 m/s. It then travels at this constant speed before decelerating uniformly at 1 m/s^2 to come to rest at station B. The total distance between the stations is 2400 meters. (a) Find the time and distance covered during the acceleration phase. (b) Find the total journey time between the two stations.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

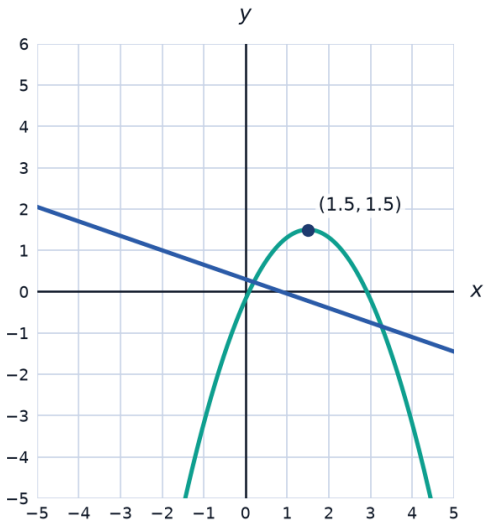
15 [Challenging] • Target: Relative and combined motion • Cross-Lesson: 8.01 Functions: Composite & Inverse

A train accelerates uniformly from rest to 30 m/s in 20 s, travels at constant speed 30 m/s for 60 s, and then decelerates uniformly to rest in 15 s. Find: (a) the acceleration in the first phase, (b) the total distance travelled, (c) the average speed for the entire journey.

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16 [Challenging] • Target: A well-designed word problem: a train between two stations • Cross-Lesson: 9.01 Circular Measure

Two particles P of mass $3\ \text{kg}$ and Q of mass $5\ \text{kg}$ are connected by a light inextensible string passing over a smooth fixed pulley. The system is held at rest and released. Taking $g = 9.8\ \text{m/s}^2$: (a) find the acceleration a of the system, (b) calculate the tension T in the string, (c) find the force exerted by the string on the pulley.



17 [Challenging] • Target: Displacement, velocity, and acceleration • Cross-Lesson: 1.01 Indices, Surds & Quadratics

A particle moves along a straight line such that its velocity in m/s at time t seconds is given by $v(t) = 6t - 2t^2$ for $0 \le t \le 4$. (a) Find the acceleration of the particle at $t = 2\ \text{s}$. (b) Determine the net displacement of the particle between $t = 0$ and $t = 4$. (c) Calculate the total distance travelled by the particle over the interval $[0, 4]$.

18 [Challenging] • Target: The suvat equations for constant acceleration • Cross-Lesson: 10.01 Series & Binomial Theorem

Practice 5.2: A particle moves in a straight line with velocity $v(t) = 3t^2 - 12t + 9\ \text{m/s}$ for $0 \le t \le 4$. (a) Find the times at which the particle is momentarily at rest. (b) Calculate the net displacement of the particle over $[0, 4]$. (c) Calculate the total distance travelled by the particle over $[0, 4]$.

19 [Challenging] • Target: Velocity-time graphs • Cross-Lesson: 12.01 Forces & Equilibrium

A block of mass $5\ \text{kg}$ is released from rest at the top of a rough plane inclined at 30° to the horizontal. The length of the incline is $8\ \text{m}$ and the coefficient of friction is $\mu = 0.2$. Using the work-energy principle and taking $g = 9.8\ \text{m/s}^2$, calculate the speed of the block at the bottom of the incline.

20 [Challenging] • Target: Motion under gravity • Cross-Lesson: 13.01 Newton's Laws

A stone is thrown vertically upwards with speed $15\ \text{m/s}$ from the top of a cliff $40\ \text{m}$ above sea level. Taking $g = 9.8\ \text{m/s}^2$ and upwards as positive: (a) find the maximum height reached above the cliff, (b) calculate the time taken for the stone to reach the sea.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Displacement, velocity, and acceleration

A particle moves so that its velocity at time t s ($t \geq 0$) is $v = t^2 - 4t$ m/s. Find when the particle is momentarily at rest.

a. $t = 3$ s (excluding $t = 0$)

b. $t = 5$ s (excluding $t = 0$)

c. $t = 0$ s only

d. $t = 4$ s (excluding $t = 0$)

22 [Foundational] • Target: The suvat equations for constant acceleration

A particle has initial velocity $u = 3$ m/s and constant acceleration $a = 3$ m/s^2. Find its speed after travelling a displacement $s = 11$ m.

a. $\sqrt{79}$

b. $\sqrt{71}$

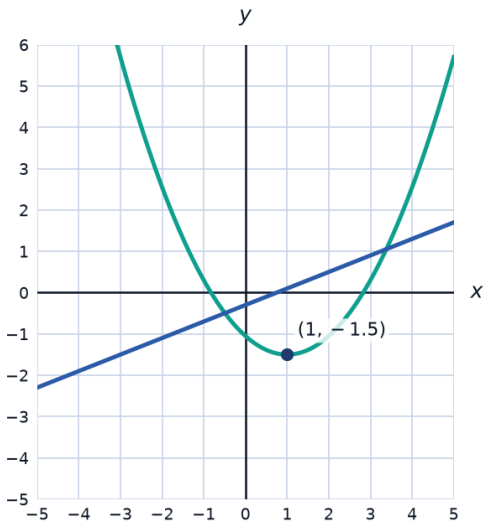
c. 75

d. $\sqrt{75}$

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23 [Foundational] • Target: Velocity-time graphs

A particle has initial velocity $u = 4$ m/s and constant acceleration $a = 4$ m/s^2. Find its displacement after $t = 6$ s.



a. 24

b. 98

c. 94

d. 96

24 [Foundational] • Target: Motion under gravity

A particle moves in a straight line such that its displacement s metres from a fixed origin after t seconds is given by the equation $s = t^3 - 6t^2 + 9t + 2$. The velocity of the particle is given by $v = \frac{ds}{dt}$. Find the value of t for which the particle is instantaneously at rest and then determine the displacement of the particle at that instant.

a. t=1 s, s=6 m

b. t=2 s, s=4 m

c. t=3 s, s=2 m

d. t=1 s, s=2 m

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25 [Foundational] • Target: Using calculus for variable acceleration

A particle of mass 2 kg is acted upon by a constant force $\mathbf{F}$. At time $t = 0$, the particle is at the origin $(0, 0)$ and has velocity $\mathbf{v} = (3\mathbf{i} + 4\mathbf{j})$ m/s. At time $t = 2$ s, the particle's velocity is $\mathbf{v} = (7\mathbf{i} + 12\mathbf{j})$ m/s. Determine the magnitude of the constant force $\mathbf{F}$ acting on the particle.

a. $\frac{5\sqrt{17}}{2}$ N

b. $5\sqrt{17}$ N

c. $\frac{5\sqrt{2}}{2}$ N

d. $5\sqrt{2}$ N

26 [Foundational] • Target: Finding when a particle is at rest or changes direction

A particle of mass 5 kg is at rest on a smooth horizontal surface. A constant horizontal force of magnitude 20 N is applied to the particle. What is the magnitude of the acceleration of the particle in m/s^2?

a. 4

b. 5

c. 20

d. 25

27 [Foundational] • Target: Relative and combined motion

A particle has initial velocity $u = 10$ m/s and constant acceleration $a = 4$ m/s^2. Find its velocity after $t = 3$ s.

a. 22

b. 21

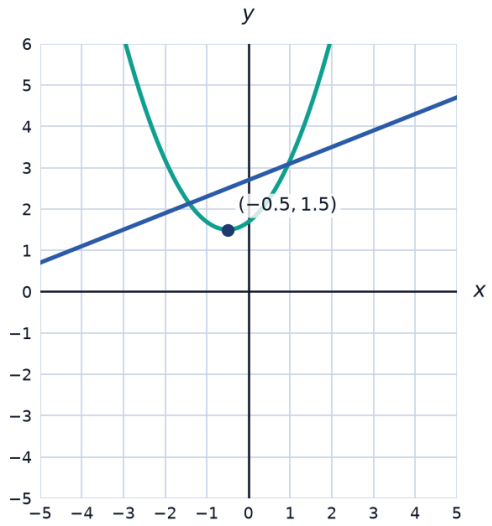
c. 23

d. 14

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28 [Foundational] • Target: A well-designed word problem: a train between two stations

A particle moves so that its velocity at time t s is $v = 8t + 3$ m/s. Find the acceleration at $t = 0$ s.



a. 13

b. 10

c. 8

d. 6

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Displacement, velocity, and acceleration

A particle moves so that its velocity at time t s is $v = 2t$ m/s. Find an expression for its displacement s in terms of t.

a. $3t^2 + C$

b. $t^2 + C$

c. 1+C

d. $2t^2 + C$

30 [Intermediate] • Target: The suvat equations for constant acceleration

A particle moves in a straight line such that its acceleration, a m s^{-2}, at time t seconds is given by $a = 6t - 18$. The particle is at instantaneous rest when $t = 1$. Given that the displacement of the particle from a fixed point O is $s = -10$ m when $t = 2$, find the displacement of the particle, in metres, when it is next at instantaneous rest.

a. -37

b. -25

c. -13

d. -5

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31 [Intermediate] • Target: Velocity-time graphs

A particle moves in a straight line such that its velocity, v m s^{-1}, at time t seconds is given by $v = 6t^2 - 18t + k$, where k is a constant. Given that the particle is momentarily at rest when $t = 2$, find the total distance traveled by the particle in the first 2 seconds.

a. 6 m

b. 4 m

c. 5 m

d. 8 m

32 [Intermediate] • Target: Motion under gravity

A particle is in equilibrium under three collinear forces: two forces of 6 N and 13 N acting in one direction, and a third force F acting in the opposite direction. Find F.

a. 19

b. 16

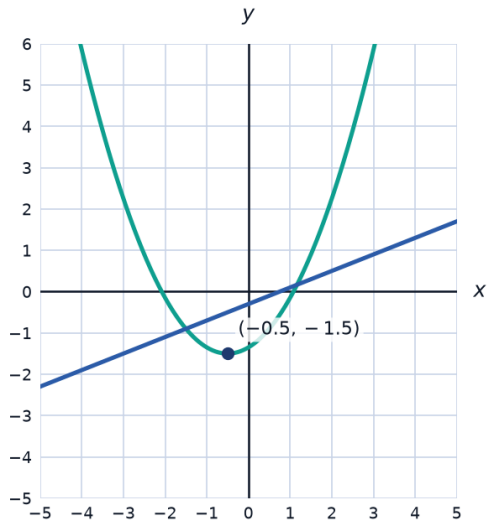
c. 21

d. 7

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33 [Intermediate] • Target: Using calculus for variable acceleration

At 08:00, a cargo vessel has position vector $(-7\mathbf{i} + 22\mathbf{j})$ km relative to a coastal radar station at the origin O, and moves with a constant velocity of $(15\mathbf{i} + 5\mathbf{j})$ km/h. At the same time (08:00), a high-speed patrol boat stationed at $(5\mathbf{i} + 6\mathbf{j})$ km sets off in a straight line at a constant speed v km/h to intercept the cargo vessel. Find the minimum speed v required for the patrol boat to intercept the cargo vessel, and the time at which interception occurs at this minimum speed.



a. Minimum speed 15 km/h, interception at 12:00

b. Minimum speed 15 km/h, interception at 10:00

c. Minimum speed $5\sqrt{10}$ km/h, interception at 12:00

d. Minimum speed 12.5 km/h, interception at 11:12

34 [Intermediate] • Target: Finding when a particle is at rest or changes direction

A supply launch S departs from a base at the origin O at time $t = 0$ to intercept an automated cargo barge C. At $t = 0$, the barge has position vector $\mathbf{r}_0 = (16; 12)$ km relative to O and maintains a constant velocity of $\mathbf{v}_C = (-15; 0)$ km/h. The launch travels in a straight line with constant velocity $\mathbf{v}_S$ and constant speed v km/h. Determine the minimum speed v required for the launch to successfully intercept the barge.

a. 9 km/h

b. 12 km/h

c. 15 km/h

d. 7.2 km/h

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35 [Intermediate] • Target: Relative and combined motion

Two particles, A and B, move in the xy-plane. Their position vectors at time t seconds, for $t \geq 0$, are given by $\mathbf{r}_A(t) = (2t;\, 4t - 1)$ and $\mathbf{r}_B(t) = (3 + 5t;\, 1 - 2t)$ respectively. The particles start their motion at $t = 0$. Find the minimum distance between particle A and particle B.

a. $\sqrt{26}$

b. $2\sqrt{13}$

c. $3\sqrt{10}$

d. $4\sqrt{5}$

36 [Intermediate] • Target: A well-designed word problem: a train between two stations

A particle moves in a straight line such that its acceleration, a m s^{-2}, at time t seconds is given by $a = 6t - 4$. When $t = 1$, the velocity of the particle is 5 m s^{-1}. Find the velocity of the particle when $t = 3$.

a. 15 m s^{-1}

b. 19 m s^{-1}

c. 21 m s^{-1}

d. 27 m s^{-1}

37 [Intermediate] • Target: Displacement, velocity, and acceleration

A particle in equilibrium is acted on by three concurrent forces. Two of the forces have equal magnitude 20 N and act symmetrically either side of the third force's line of action. If the angle between each of the two equal forces and the third force is 120°, find the magnitude of the third force.

a. 18

b. 20

c. 25

d. 22

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The suvat equations for constant acceleration • Cross-Lesson: 10.01 Series & Binomial Theorem

A particle of mass 6 kg moving at 6 m/s collides and sticks to a particle of mass 6 kg moving at 0 m/s (same line of motion). Find the common velocity after collision.

a. 2

b. 4

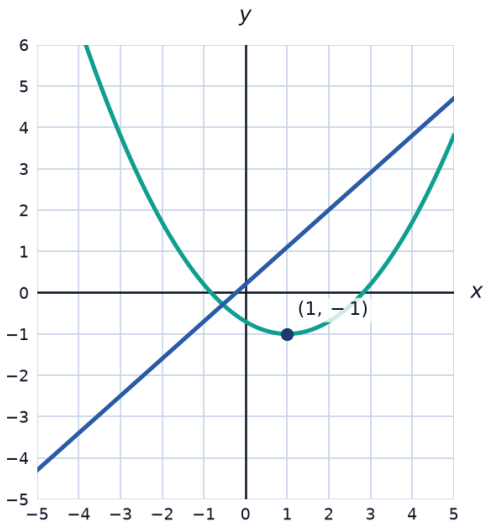
c. 6

d. 3

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39 [Challenging] • Target: Velocity-time graphs • Cross-Lesson: 12.01 Forces & Equilibrium

Particles of mass 2 kg and 5 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.



a. 4.20

b. 8.40

c. 4.70

d. 3.70

40 [Challenging] • Target: Motion under gravity • Cross-Lesson: 13.01 Newton's Laws

A particle accelerates uniformly from $u = 2$ m/s to $v = 14$ m/s in $t = 6$ s. Find its acceleration.

a. 1

b. 3

c. 2

d. 12/7

41 [Challenging] • Target: Using calculus for variable acceleration • Cross-Lesson: 14.01 Energy, Work & Power

Relative to a fixed origin O, ship A has an initial position vector of $(24\mathbf{i} + 10\mathbf{j})$ km and moves with a constant velocity of $(-2\mathbf{i} + 5\mathbf{j})$ km h^{-1}. Ship B has an initial position vector of $(-20\mathbf{i} - 18\mathbf{j})$ km and moves with a constant velocity of $(u\mathbf{i} + 12\mathbf{j})$ km h^{-1}, where u is a constant. Given that ship B intercepts ship A, find the speed of ship B.

a. 15 km h^{-1}

b. 9 km h^{-1}

c. 21 km h^{-1}

d. 34 km h^{-1}

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42 [Challenging] • Target: Finding when a particle is at rest or changes direction • Cross-Lesson: 15.01 Representation of Data

At 14:00, a coastguard radar station detects two vessels, A and B. Relative to the radar station, vessel A has position vector $(-4\mathbf{i} + 6\mathbf{j})$ km and moves with a constant velocity of $(8\mathbf{i} + 5\mathbf{j})$ km h^{-1}. Vessel B has position vector $(20\mathbf{i} - 4\mathbf{j})$ km and moves with a constant velocity of $(u\mathbf{i} + 10\mathbf{j})$ km h^{-1}, where u is a constant. Given that vessel B intercepts vessel A, find the value of u.

a. $u = -4$

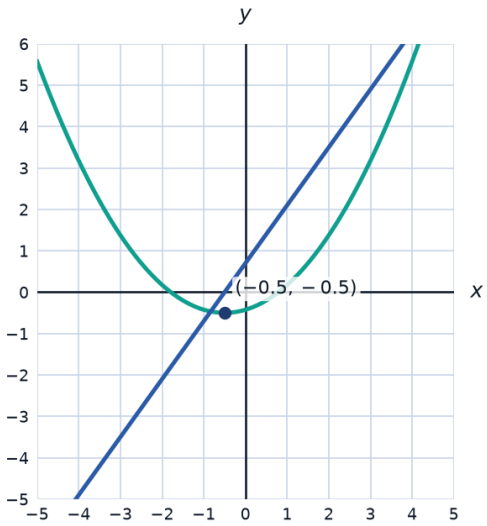
b. $u = 4$

c. $u = -2$

d. $u = -8$

43 [Challenging] • Target: Relative and combined motion • Cross-Lesson: 16.01 Permutations & Combinations

A particle moves in a straight line with acceleration $a(t) = 6t - 8$ m/s^2. If it starts from rest at the origin (at $t = 0$), at what time $t > 0$ does the particle momentarily come to rest again?



a. $t = 4$ s

b. $t = \frac{4}{3}$ s

c. $t = \frac{8}{3}$ s

d. $t = 8$ s

44 [Challenging] • Target: A well-designed word problem: a train between two stations • Cross-Lesson: 17.01 Discrete Random Variables

A constant force of 15 N is applied to an object, causing it to move a distance of 4 m in the direction of the force. Calculate the work done by the force.

a. 60 J

b. 19 J

c. 11 J

d. 3.75 J

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45 [Challenging] • Target: Displacement, velocity, and acceleration • Cross-Lesson: 18.01 Binomial Distribution

Three concurrent forces keep a particle in equilibrium, each pair separated by 120°, and two of the forces have magnitude 28 N. Find the magnitude of the third force.

a. 26

b. 30

c. 33

d. 28

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